

Predictive Foveal Isometry

A conditional mean-embedding theory and the PIVOT self-supervised learning mechanism

Research Proposal - July 2026

Core thesis

A useful visual token should directly estimate the conditional mean embedding of withheld natural context under a declared foveal observation channel. This gives the deployed backbone an explicit predictive kernel, while verified geometric transport makes that kernel spatially usable.

Research target

Test whether this representation principle can outperform the original DINO under matched resources and compete with DINOv2-style and JEPA-family objectives, with the strongest expected gains on dense transfer, retrieval, few-shot organization, and robustness.

Status of the claims

This is a candidate theory and experimental program, not an empirical result. The semantic assumption, finite-dimensional approximation, target construction, and scale-up decisions are tied to explicit falsification gates.

0. Abstract

Predictive Foveal Isometry (PFI) proposes that a deployed visual token should encode the predictive state induced by a local visual event: the conditional distribution of natural context that remains withheld from that event. The PIVOT mechanism realizes this principle through direct conditional mean-embedding regression. A separately trained and frozen context tokenizer maps withheld contexts to stable descriptors. Fixed random Fourier features approximate characteristic kernels over those descriptors. The backbone is trained directly, without a learned representation-defining projector, to regress sampled context features. By the conditional-expectation projection theorem, the population optimum is the conditional mean embedding of the context law. Consequently, token inner products implement an explicit predictive kernel rather than an unexplained geometry that emerges only through a training head. Multiple context channels represent near, far, global, appearance, and temporal predictive relations. Known crop and warp geometry supplies an overlap-aware transport loss, while a temporary variance floor protects early optimization without defining the representation distribution. Region and image descriptors are fixed poolings of the canonical token field. The decisive empirical question is whether held-out predictive-state distortion at the retained backbone predicts and improves global and dense transfer beyond compute-matched DINO, DINOv2-style, VICReg, I-JEPA, and LeJEPA baselines.

0.1. Standalone statement of contribution

PIVOT makes three commitments that jointly distinguish it from neighboring approaches:

- **Semantic commitment:** similarity is defined by equality or proximity of conditional laws over declared withheld context, not by a universal marginal distribution for embeddings.
- **Representation commitment:** the defining prediction loss acts on retained backbone token blocks. Training-only heads cannot absorb the property that is claimed at deployment.
- **Spatial commitment:** semantic prediction and geometric transport are separate constraints. Predictive equivalence does not, by itself, guarantee stable localization.

0.2. Document map

Section	Purpose
1-2	Define the scientific question, task family, and representation requirements.
3-4	Specify the predictive-state construction and state the theoretical results and limits.
5-6	Define the training objective, architecture, sampling, and compute accounting.
7	Position PIVOT against DINO, DINOv2/v3, VICReg, JEPA, and LeJEPA.
8-10	State falsifiable predictions, decisive experiments, and kill tests.
11-12	State novelty, success criteria, and the bounded conclusion.

1. Problem and design objective

Self-supervised visual learning has produced powerful transferable features through view agreement, redundancy reduction, reconstruction, clustering, and latent prediction. DINO learns teacher-student prototype assignments across augmented crops. DINOv2 combines that objective family with extensive data curation, regularization, distillation, and scale. DINOv3 retains multi-crop self-distillation and adds Gram anchoring to protect dense feature geometry during long, large-model training. JEPA-family methods predict withheld latent content. VICReg and LeJEPA add explicit output-distribution constraints that prevent collapse or improve estimator conditioning. These systems are strong, but they do not answer the same question as PIVOT: what should the inner product between two deployed visual tokens mean?

PIVOT proposes predictive equivalence. Two visual events should be close when they imply similar conditional laws over withheld natural context. A brown curved edge on fur, a wheel fragment, a face part, or a repeated architectural element can be locally ambiguous; its typical surroundings help resolve its identity. Conversely, a random, contextually silent mark should not become a major semantic distinction merely because a reconstruction objective can preserve it.

Flagship claim

For mainstream recognition, retrieval, and dense-transfer tasks, the conditional law of multichannel natural context contains a useful semantic geometry. A retained visual token that directly estimates this predictive state should organize unseen examples more usefully than a representation whose geometry is induced only indirectly by augmentations, marginal regularizers, or a training predictor.

1.1. Scope

The scientific objective is an objective-level comparison at controlled architecture, pretraining data, input resolution, token count, optimization schedule, hyperparameter-search budget, and total compute. Full-system comparisons must count the cost of the context tokenizer and target evaluation. The proposal does not claim immediate superiority to every large foundation-model system, whose performance also reflects data curation, model scale, distillation, and extensive engineering.

The strongest prior is for semantic correspondence, segmentation, keypoint transfer, retrieval calibration, few-shot organization, and robustness when natural predictive relations remain stable under shift. Image-level linear probing is important but is not the only or necessarily the first expected win.

1.2. Two motivating gaps in related work

First, collapse prevention and representation semantics should not be conflated. A variance floor, covariance penalty, whitening term, or Gaussian-matching term can be valuable as an optimization or conditioning device. It does not by itself determine what information a general-purpose representation should encode. PIVOT therefore treats marginal distribution shaping as non-semantic unless a downstream task model and its assumptions are stated explicitly.

Second, the exceptional performance of DINO-family systems is not fully predicted by their loss values or a simple mechanistic reading. Their local-to-global crop matching deliberately creates a difficult and sometimes noisy relation: a small crop is trained toward a teacher distribution derived from a global crop of the same image even when the small crop does not visibly contain the dominant object or scene evidence. Teacher sharpening, centering, EMA dynamics, and multi-crop pressure are mechanistically understandable, but the magnitude and breadth of transfer remain an empirical achievement rather than a consequence of a sufficiently predictive theory. PIVOT is designed so that the target relation and its failure can be measured directly.

2. Task family and representation requirements

Let an image arise from entities, parts, geometry, attributes, materials, depth, and sparse relations. The target family contains tasks whose outputs attach to images, regions, locations, or pairs of regions and whose regression functions vary smoothly with an appropriate predictive-state distance. This includes classification, retrieval, detection, semantic and instance segmentation, correspondence, keypoints, depth, and moderate distribution shift.

The family excludes labels that are invisible to every declared context channel. A random serial number on an otherwise uninformative patch may be important for OCR but predictively silent under the chosen channels. This is a substantive boundary of the hypothesis.

Requirement	Failure prevented	Operational test
Predictive cross-image geometry	Poor retrieval, kNN organization, and few-shot neighborhoods	Held-out predictive distortion should rank transfer after controlling for feature rank, model size, and compute.
Foveal contextual sufficiency	Ambiguous patches cannot support dense labels	A graded local-surround channel should outperform local-only and unrestricted-global extremes.
No observation leakage	The token predicts context because it can directly see the target	The backbone input and attention mask must expose exactly the declared event channel; withheld regions remain inaccessible.
Spatial transport	Good semantics but unstable localization and correspondence	Known-warp residual should predict dense alignment and fall when transport is trained.
Direct backbone target	A head absorbs the SSL property	Removing all training-only objects must leave the defining predictive geometry at retained tokens.
Nondegenerate optimization	Transient constant or low-variance features	Feature variance and effective rank remain above warm-up floors without permanent marginal shaping.
Pooling coherence	Global and dense interfaces drift apart	Fixed pooled descriptors retain useful image and region transfer without a separate incompatible geometry.
Finite-sample generalization	A high-capacity model memorizes realized contexts rather than a conditional law	Held-out images, context resampling, capacity sweeps, and repeated-event synthetic tests separate regression generalization from memorization.

3. Predictive Foveal Mean-Embedding Isometry

3.1. Declared foveal event channel

For image x and spatial location p , define a deterministic event channel Ψ . It returns a sharp central observation, progressively degraded surrounding rings, relative position information, a scale tag, and a visibility mask. The backbone token used in the defining loss must be a function only of this event channel; otherwise the population target changes from a conditional law given the event to a conditional law given a richer, leaking input.

$$e_p = \Psi(x, p) = [\text{center_high}, \text{ring_1}, \dots, \text{ring_L}, \text{relative_position}, \text{scale}, \text{visibility}]$$

The attenuation profile is a declared task prior. It sets the balance between locality and contextual disambiguation. The first implementation can realize it with masked and multiresolution crops; a native architecture is optional.

3.2. Withheld context channels

Let Ω_m define context channel m . Examples include a near annulus, a distant crop, a low-resolution whole-image view, an alternative appearance view, or a temporal continuation.

$$C_m = \Omega_m(X, p)$$

Multiple channels are preferable to one universal context because they expose different predictive relationships and reduce dependence on a single observational choice. Each channel occupies a declared token subspace or contributes through a fixed weighted concatenation. Channels must be audited for object absence, positional shortcuts, and nuisance correlations.

3.3. Frozen context descriptors and characteristic features

A context tokenizer T_m is trained separately and frozen. Its role is to retain a stable multiscale description of withheld context, not to co-adapt with PIVOT. For $y_m = T_m(C_m)$, a fixed random Fourier feature map approximates a Gaussian kernel:

$$\phi_m(y) = \sqrt{2/D_m} [\cos(w_1^T y + b_1), \dots, \cos(w_{D_m}^T y + b_{D_m})]$$

Several bandwidths can resolve coarse and fine differences. In the infinite-feature limit, a characteristic kernel makes the mean embedding injective over distributions of the frozen descriptor. At practical token dimensions, PIVOT learns a finite sketch; approximation quality and spectrum must be measured rather than assumed.

3.4. Predictive state and kernel

$$\mu_m(e) = E[\phi_m(T_m(C_m)) \mid E=e]$$

$$\mu(e) = \text{direct_sum}_m \sqrt{\alpha_m} \mu_m(e), \alpha_m \geq 0, \sum_m \alpha_m = 1$$

$$K_\mu(e, e') = \langle \mu(e), \mu(e') \rangle$$

The kernel has a useful probabilistic interpretation. For independent C drawn from $P(C|e)$ and C' drawn from $P(C|e')$,

$$K_\mu(e, e') = E[k(T(C), T(C')) \mid e, e']$$

Thus two events are similar when their possible withheld contexts are similar under the declared context kernel. This is a property of conditional laws, not of an imposed marginal distribution over backbone features.

3.5. Deployed representation property

Let $h_p(x)$ be the retained final token, divided into the same channel blocks as the target sketch. PFI with distortion epsilon requires:

$$E \|h_p(x) - \mu(\Psi(x, p))\|^2 \leq \epsilon$$

Cosine normalization is applied only at an evaluation interface when needed. The unnormalized norm is retained because, for bounded kernels, it measures concentration of the conditional context law at the chosen bandwidths. It may correlate with ambiguity, but it is not automatically a calibrated probability or epistemic confidence score.

PIVOT separates semantic target construction from the deployed representation.

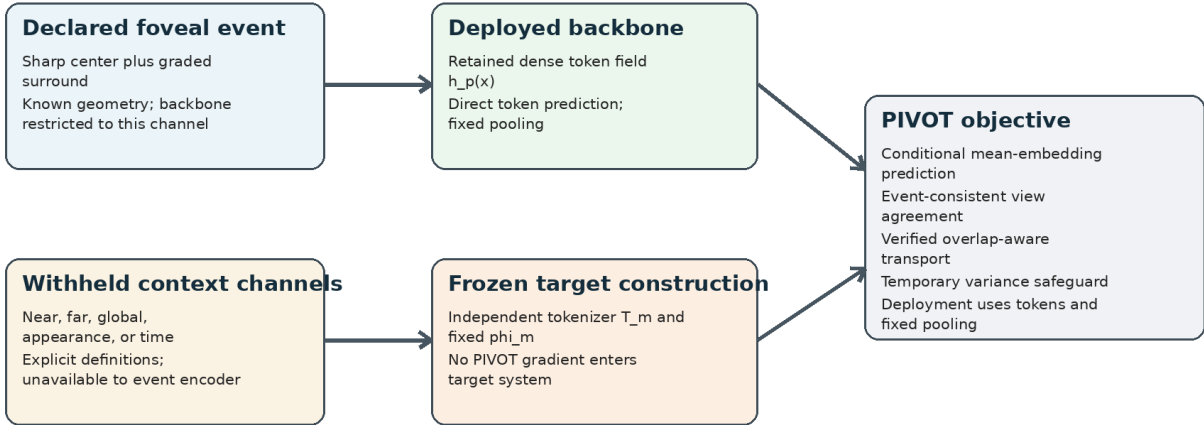


Figure 1. Semantic target construction is frozen and separate from the deployed backbone. The defining regression acts directly on retained tokens.

4. Formal justification and theoretical limits

4.1. Theorem 1 - direct regression identifies the predictive state

For any square-integrable target $Z = \phi(T(C))$, consider the population objective

$$L(f) = E \|f(E) - Z\|^2$$

The unique minimizer almost surely is

$$f^*(e) = E[Z \mid E=e] = \mu(e)$$

This is the orthogonal projection of Z onto the space of square-integrable functions of E . Ordinary vector regression from event-context pairs therefore identifies the conditional mean embedding at the population optimum. No pairwise density-ratio critic is required.

4.2. Corollary - the retained Gram realizes a predictive kernel

$$\langle h^*(e), h^*(e') \rangle = \langle \mu(e), \mu(e') \rangle = K_{\mu}(e, e')$$

The cross-image geometry is consequently tied to a declared target at the backbone optimum. In contrast to methods whose defining relation is applied only after a learned projector, PIVOT makes the property directly testable on retained tokens.

4.3. Conditional-law separation

If ϕ is the exact feature map of a characteristic kernel, then equality of conditional mean embeddings is equivalent to equality of conditional laws over $T(C)$. The mathematical statement concerns the descriptor distribution itself. Whether $T(C)$ is sufficient for the intended visual semantics is a separate empirical assumption. Finite random features weaken exact separation to an approximation.

4.4. Task consequence

If a downstream regression function is Lipschitz in predictive-state distance and the training set covers the relevant state space, then small PFI distortion controls local neighborhood distortion and supports kNN, kernel methods, retrieval, and shallow adaptation. This is a conditional transfer statement. It does not establish universal linear separability, perfect reconstruction, or superiority on labels that are silent under the declared context channels.

4.5. Why this is not a theorem about the universal marginal distribution of representations

The theory fixes the conditional target $\mu(e)$. It does not require the population distribution of $h(E)$ to be spherical, Gaussian, whitened, uniform, or equal-variance in every direction. Any anisotropy that follows from the predictive task is allowed. The temporary variance term in Section 5 is only an optimization safeguard. Permanent covariance, whitening, or Gaussian-matching penalties are optional ablations and must be judged by whether they improve predictive distortion and transfer rather than by an assertion that one marginal law is universally semantic.

4.6. Main gaps between the population theorem and a working system

Gap	Why the theorem does not solve it	Required evidence
Finite-sample regression	Each natural event is usually observed with one realized context; an overparameterized network can memorize training pairs.	Held-out context prediction, repeated-event synthetic data, augmentation-based context resampling, and capacity/regularization sweeps.
Observation leakage	If h sees the withheld region, the conditional variable is not the declared event.	Hard masks or attention restrictions verified by intervention and leakage probes.
Target semantics	The theorem is exact relative to T and Ω , even if they encode spurious cues.	Multiple target tokenizers, nuisance interventions, target-only probes, and domain-held-out audits.
Finite sketch	A small token cannot preserve an arbitrarily rich characteristic embedding.	Dimension/bandwidth sweeps, spectrum diagnostics, and approximation error against larger offline sketches.
Optimization	The population optimum may be difficult to reach with noisy targets.	Loss decomposition, gradient-noise measurements, target averaging, curriculum, and seed variance.
Transport compatibility	The predictive state can legitimately change when a crop or transform changes visible evidence.	Apply transport only to overlap-valid events and context channels whose conditional target is preserved under the mapped transform.

5. PIVOT training mechanism

5.1. Deployed and training-only objects

Status	Objects
Deployed	Vision backbone token field $h_p(x)$, optional global token, retained normalization interface, and fixed pooling operators.
Frozen target system	Independent context tokenizer T_m , target standardization, fixed random Fourier features ϕ_m , channel weights, masks, and warp maps.
Training-only sampling	Foveal event sampler Ψ , context samplers Ω_m , crop correspondences, overlap masks, target repetitions, and running optimization statistics.
Explicitly absent from the defining path	No learned projector or predictor between a retained token block and its predictive target; no target gradient into the frozen context system.
Optional safeguards	A temporary variance floor, gradient clipping, and target averaging. These do not define semantic geometry.

5.2. Target construction

1. Train and freeze one or more multiscale context tokenizers using reconstruction-oriented or information-preserving objectives. Report the target-only transfer performance so that hidden teacher distillation is visible.
2. Standardize each context channel using frozen statistics estimated on a held-out stream.
3. Sample fixed random features at several RBF bandwidths. Allocate a known output block to each channel-bandwidth pair and verify that the total sketch fits the retained token dimension.
4. Stop gradients at the target path. PIVOT training never changes T_m , ϕ_m , target normalization, or context sampling definitions.
5. Audit target sufficiency and shortcuts. Probe appearance, structure, layout, position, background, acquisition source, and object-presence signals before backbone scale-up.
6. Build a larger offline target sketch for diagnostics. Measure how well the practical token-dimensional sketch preserves its neighborhoods and conditional-law distinctions.

5.3. Main predictive loss

For event e and one or more sampled contexts from each channel, define $z_m = \phi_m(T_m(c_m))$. The retained token block $h_m(e)$ is trained directly:

$$L_{\text{pred}} = \sum_m \alpha_m E \rho(h_m(e) - \text{stopgrad}(z_m))$$

The stochastic target is not the predictive state itself; its conditional expectation is. Multiple target samples, alternative masks, repeated crop templates, and batch-level grouping of similar event observations can reduce variance. The critical validation is held-out conditional prediction, not low training error alone.

5.4. Event-consistent cross-view agreement

When two transformed views retain the same declared event information, their corresponding tokens should agree:

$$L_{\text{view}} = E \|h(e^{(1)}) - h(e^{(2)})\|^2$$

This term is only applied when the event channels are equivalent under the transformation. It must not recreate an unconditional DINO-style rule that every local crop should share the same global semantic target. If a crop removes predictive evidence, PIVOT permits the state to change.

5.5. Verified geometric transport

Known crop, resize, flip, and mild warp maps determine a token reindexing operator W_g and an overlap mask M_g :

$$L_{\text{transport}} = E \rho(M_g * [h(gx) - W_g h(x)])$$

The term is verified rather than declared universally exact. Interpolation, patchification, occlusion, changed foveal surrounds, and non-equivariant target descriptors produce measurable residuals. Transport is applied only where the transformed event and relevant context channels satisfy a declared compatibility condition.

5.6. Warm-up nondegeneracy

$$L_{\text{var}} = \sum_j [\gamma - \text{Std}_{\text{batch}}(h_j)]^2$$

A nonconstant frozen target excludes a constant function at the population optimum, but early optimization can still enter low-variance states. The floor is annealed after predictive variance and effective rank stabilize. Full whitening, unit-norm regression, covariance diagonalization, and isotropic-Gaussian matching are not defining assumptions.

5.7. Final objective

$$L_{\text{PIVOT}} = L_{\text{pred}} + \lambda_{\text{view}} L_{\text{view}} + \lambda_{\text{T}} L_{\text{transport}} + \lambda_{\text{var}} L_{\text{var}}$$

This four-term objective is the primary implementation. Fixed pooled prediction may be added for a global token. Edited-state or marginal-distribution auxiliaries are added only when a measured diagnostic failure predicts downstream performance.

6. Architecture, sampling, and compute

6.1. Minimum viable implementation

Use an ordinary ViT so that the objective can be tested without architectural novelty. Construct a foveal input from a sharp local crop and one or more blurred or downsampled surrounds. Enforce the declared channel with explicit masks or attention restrictions. Obtain corresponding dense tokens by known crop maps. This tests the central theory with minimal architectural confounding.

6.2. Optional native foveal architecture

After objective-level evidence, a native model can expose the channel more efficiently: full-resolution local tokens, pooled intermediate-range tokens, a small global low-resolution grid, and distance-dependent attention bias. Relative geometry should be separated from semantic content so that absolute dataset position cannot become an easy shortcut.

6.3. Sampling curriculum

- Early: short-range contexts, high overlap, moderate augmentations, repeated target views, and stable low-frequency target blocks.
- Middle: distant crops, low-resolution global context, larger scale variation, and stronger object-absence audits.
- Late: ambiguous local events, repeated structures, occlusion, instance-discriminating regions, and high-resolution transport.
- Hard sampling: prioritize high held-out predictive residual or disagreement across context channels, not similarity mining from the current backbone alone.
- Null and mismatch sampling: include events whose local crop lacks the dominant object, and test whether the representation expresses ambiguity instead of false global certainty.
- Edited states: reserve a controlled minority for response diagnostics and targeted repair.

6.4. Computational profile and matched-budget accounting

Each event requires one backbone forward and several frozen target evaluations. There is no quadratic pairwise critic. Random features are matrix operations on cached descriptors, and target blocks can be precomputed for repeated crop templates. Nevertheless, the target tokenizer is real compute and must not be hidden.

Accounting item	Matched-budget rule
Backbone pretraining	Same architecture, input tokens, image count, optimizer family, schedule length, and measured training FLOPs.

Target tokenizer	Report its pretraining images and FLOPs. Provide both full-system cost and amortized cost when reused across runs.
Target inference	Count frozen context forward passes, cache construction, storage, and data-loading overhead.
Hyperparameter search	Match tuning trials or report a fixed preregistered search space for every method.
Augmentation and sampling	Match pixel exposure and token exposure; do not give PIVOT extra views without charging their cost.
Evaluation	Use identical frozen/fine-tuned heads, schedules, seeds, and checkpoint-selection information.

7. Mechanistic position relative to powerful baselines

7.1. VICReg and LeJEPa: collapse control versus a semantic prior

VICReg combines invariance with variance and covariance regularization. Its variance and covariance terms are naturally interpreted as anti-collapse and redundancy-control mechanisms; they shape optimization and conditioning but do not identify the semantic content of the representation. PIVOT is compatible with using such terms temporarily, provided they are not treated as the answer to which distinctions the representation should preserve.

LeJEPa makes a stronger claim: under specified downstream estimator families, task classes, covariance constraints, and risk criteria, an isotropic Gaussian embedding distribution is optimal. That can be a meaningful theorem about worst-case probe estimation or preconditioning. It does not by itself establish that the representation contains the right visual information, nor that an isotropic Gaussian is a task-independent semantic law of nature. PIVOT therefore separates two questions: what the token predicts is defined by the conditional context law; how the marginal token cloud is stabilized is secondary and empirically testable.

PIVOT design rule

No permanent marginal distribution prior is allowed to define semantic geometry in the primary method. A VICReg-, whitening-, or SIGReg-style term may be tested as an optimizer aid, but it must improve held-out predictive distortion or downstream transfer without erasing predictive anisotropy.

7.2. DINO, DINOv2, and DINOv3: local-to-global pressure and unexplained performance

DINO passes all crops through the student and only global crops through the EMA teacher, creating local-to-global correspondences. This is a powerful source of invariance and semantic grouping, but it also asks a local crop to match a global teacher assignment even when the local crop lacks the object or scene evidence that dominates the global view. The resulting target can be viewed as structured label noise, contextual completion pressure, or a regularized clustering signal. The mechanism explains the direction of the tension; it does not make the final level of performance predictable from the loss alone.

DINOv2 demonstrates that data curation, scale, regularization, masked patch prediction, architecture choices, and distillation can turn this family into a strong general-purpose system. DINOv3 further shows that dense feature quality can degrade during long training even while global objectives improve, and introduces Gram anchoring to preserve pairwise patch geometry using an earlier model state. These developments reinforce the proposal's central methodological demand: global and dense geometry should be measured at the retained tokens, and a training loss should be judged by whether its own held-out metric predicts transfer.

7.3. How PIVOT addresses the local-crop mismatch

DINO-family tension	PIVOT response	Remaining risk
A local crop may not contain the dominant object but is matched to a global teacher assignment.	The target is a conditional context state. An ambiguous or object-absent event should approach the average embedded law over compatible contexts, not a confident instance-specific global prototype.	With one realized context per event, finite-sample memorization can still create false certainty. Target repetition, held-out prediction, and capacity controls are essential.

The crop relation is implicit in augmentation sampling.	Psi and Omega explicitly state what is visible and withheld, including scale, overlap, and object-presence audits.	A poor declaration can encode the wrong semantics or shortcut cues.
Training loss is not a reliable direct measure of feature quality.	Held-out predictive residual, kernel distortion, transport residual, and channel-specific errors are preregistered model-selection metrics.	They may still fail to correlate with downstream tasks; that is a central refutation.
Global objectives can degrade dense maps.	Dense retained tokens are the primary prediction objects, and transport is separately constrained.	Direct random-feature regression may consume token capacity or conflict with global abstraction.

7.4. Mechanistic comparison

Method	What defines semantics	Marginal prior / collapse control	Main interpretive limitation
DINO	Agreement with EMA-teacher prototype distributions across views and crops.	Centering, sharpening, EMA teacher, and view design.	Strong emergent geometry, but local-to-global targets and performance are not fully explained by a direct deployed-token statistic.
DINOv2/v3	DINO/iBOT-style objectives plus scale, curation, and system refinements; DINOv3 adds Gram anchoring for dense geometry.	Extensive stabilization, regularization, distillation, and anchoring.	System success mixes objective quality with data and engineering; dense and global pressures can conflict.
VICReg	View invariance.	Variance and covariance penalties.	Anti-collapse structure does not specify which visual distinctions should be represented.
I-JEPA	Prediction of withheld target-encoder latents.	Target encoder, predictor, masking, and stop-gradient design.	The target representation and predictor determine or absorb part of the geometry.
LeJEPA	JEPA prediction plus a prescribed isotropic Gaussian marginal.	SIGReg Gaussian matching.	The optimality theorem is conditional on task/probe/risk assumptions and does not establish semantic sufficiency.
PIVOT	Conditional mean embedding of independently frozen multichannel context descriptors.	Nonconstant targets plus a temporary variance floor.	Success depends on the event/context declaration, tokenizer sufficiency, finite sketch, and generalization to conditional means.

7.5. Why PIVOT could win

- Every spatial event receives structured prediction targets rather than only a small number of image-level assignment constraints.
- Ambiguity is representable: conditional mean magnitude and channel disagreement can express that a local event does not determine one global object.
- Dense tokens are the primary object, aligning the pretext task with segmentation, correspondence, detection, and keypoint evaluation.
- The retained representation has explicit held-out distortion metrics that can support unlabeled checkpoint and channel selection.
- Known geometry supervises spatial transport rather than relying entirely on attention and crop consistency.
- The method does not require the semantic state to obey a universal Gaussian, whitening, or uniformity prior.

7.6. Why PIVOT may lose

- DINO's noisy local-to-global pressure may be a feature rather than a bug: it can force object-level abstraction more aggressively than conditional regression.
- A frozen tokenizer may import the wrong invariances, become an expensive hidden teacher, or be too weak to define useful context laws.
- Direct loss on a limited token dimension can overconstrain the backbone; a learned projection head may in practice protect useful information.
- Conditional mean embeddings can smooth away rare instance details that recognition or detection needs.
- The target noise and context-forward cost can erase gains under a genuinely matched full-system budget.

- DINOv2/v3 performance is partly a data-and-engineering frontier; a cleaner objective alone is unlikely to overcome a major scale gap.

8. Falsifiable predictions

7. Predictive distortion ranks transfer. Held-out token prediction error and predictive-kernel distortion should explain kNN, retrieval, few-shot, and dense transfer better than the respective training losses from matched DINO, VICReg, or JEPA runs.
8. Foveal context has a nonmonotonic optimum. Local-only events should underperform on ambiguous dense labels; unrestricted high-resolution global input should weaken locality or create leakage; graded surround should dominate both extremes.
9. Object-absent local crops should express ambiguity. Their token norms, channel disagreement, or predictive entropy proxies should differ from object-present crops rather than being forced into the same confident global assignment.
10. Distributional features matter. Random Fourier features of frozen context descriptors should outperform direct raw-descriptor regression when conditional contexts are multimodal, but not necessarily when they are approximately unimodal.
11. Transport remains complementary. Removing $L_{\text{transport}}$ should leave context prediction nearly unchanged while degrading correspondence, segmentation boundaries, and keypoint transfer.
12. Token-direct training matters. Adding a learned predictor should lower training loss but may move the defining predictive geometry away from retained tokens; the prediction must be evaluated both before and after the predictor.
13. Multiple context channels outperform one. Near, far, and global blocks should provide complementary transfer gains and reduce sensitivity to any single context definition.
14. Marginal shaping is secondary. Permanent covariance, whitening, or Gaussian-matching terms should not improve transfer unless they also improve held-out predictive geometry or optimization stability; if they help independently, the proposal must acknowledge that PIVOT is incomplete.
15. Norm measures conditional concentration, not universal confidence. It should correlate with multimodality or unpredictability only after controlling for kernel bandwidth and target scale.
16. Context-silent tasks remain a boundary. PIVOT should lose to reconstructive methods on randomized marks, exact OCR details, and labels invisible to declared contexts.
17. Central refutation. If PIVOT fits well-audited targets at retained tokens but its predictive metrics neither rank nor improve broad transfer against compute- and rank-matched baselines, the predictive-state mechanism is weakened regardless of formal elegance.

9. Decisive experimental program

E0. Target-system pilot before backbone training

- Train small independent context tokenizers on synthetic data, CIFAR-100, ImageNet-100, and an ImageNet-1k subset.
- Audit object presence, position, background, camera/source, texture, shape, layout, and acquisition cues in each target channel.
- Test random-feature approximation across dimensions and RBF bandwidths, including a large offline reference sketch.
- Verify that empirical mean embeddings distinguish known multimodal conditional laws in controlled scenes.
- Measure effective rank and spectral decay at dimensions matching ViT-S and ViT-B token budgets.
- Reject target settings that are nearly constant, nearly orthogonal, dominated by absolute position, or strong enough to make the experiment ordinary teacher distillation.

E1. Frozen checkpoint zoo

Measure PIVOT prediction residual, approximate predictive-kernel distortion, transport error, norm concentration, and effective rank at multiple layers of public supervised, DINO, DINOv2, DINOv3, MAE, VICReg, I-JEPA, and

LeJEPa checkpoints. This is an early test of whether the proposed metric tracks known transfer quality before training a PIVOT backbone. Because architectures and data differ, the result is diagnostic rather than causal.

E2. Controlled repeated-event task family

Generate objects, parts, positions, depth, lighting, materials, backgrounds, occlusion, contact, randomized colors, object-absent crops, and context-silent factors. Repeated event types and interventions permit direct estimation of conditional laws. Separate target-system error, representation error, finite-sample memorization, and downstream-task mismatch.

E3. Matched training ladder

Stage	Dataset / model	Purpose	Scale-up gate
A	CIFAR-100, small ViT or ConvNet	Debug event construction, target noise, and optimization. Not a decisive foveal-semantic benchmark because spatial resolution is limited.	Stable held-out prediction; no leakage; competitive recognition at matched compute.
B	ImageNet-100, ViT-S/16	First decisive objective comparison and full ablation grid.	Match DINO global transfer within run variance and obtain a clear dense/retrieval or robustness advantage.
C	ImageNet-1k, ViT-S/16	Test generality and robust checkpoint ranking.	Broad Pareto improvement over original DINO or a compelling task-specific advantage with honest cost.
D	ImageNet-1k, ViT-B/16	Test scale and token-dimensional bottlenecks.	Benefits persist after full-system compute accounting and target dimension sweeps.
E	Larger curated data / DINOv2-style recipe	System-level comparison, only after objective evidence.	Mechanism survives matched data, resolution, distillation, and engineering controls.

Primary baselines are original DINO, a feasible DINOv2-style recipe, DINOv3-style Gram anchoring where reproducible at the chosen scale, I-JEPa, LeJEPa, MAE, VICReg, and direct regression of raw frozen context descriptors. The target-tokenizer cost is reported both amortized and in full.

E4. Evaluation suite

Group	Measurements
Recognition	CIFAR-100, ImageNet-100, and ImageNet-1k kNN and linear probes; 1/5/10-shot adaptation; calibrated and open-set classification.
Retrieval	Object and instance retrieval, landmark retrieval, nearest-neighbor calibration, and background-controlled retrieval.
Dense	COCO detection and instance segmentation, ADE20K semantic segmentation, semantic correspondence, keypoint transfer, and NYUv2 depth.
Robustness	ImageNet-C/R/A/Sketch, background replacement, camera/compression shift, scale shift, and combined nuisances.
Controlled probes	Parameter-matched two-layer MLP, kernel ridge with fixed effective degrees of freedom, shallow attention heads, and token-only versus pooled interfaces.
Direct metrics	Held-out prediction residual, large-sketch kernel distortion, transport error, feature variance, effective rank, norm concentration, object-absence response, channel disagreement, and intermediate-layer peaks.

E5. Mechanism-separating ablations

- Conditional mean embedding versus direct raw-descriptor regression and pairwise affinity estimation.

- Frozen independent target versus random target, EMA target, jointly learned target, and semantic teacher target.
- No projector versus learned predictor versus fixed orthogonal compression.
- Local-only, unrestricted-global, and graded foveal event channels with explicit leakage probes.
- Single context channel versus near + far + global blocks.
- Object-present versus object-absent crop sampling and predictability-based channel weighting.
- No transport versus token transport versus relational Gram transport.
- Temporary variance floor versus permanent VICReg covariance/variance versus LeJEPa-style SIGReg versus no safeguard.
- Fixed pooled descriptors versus separately optimized global representations.
- One target sample versus target averaging, repeated masks, and grouped similar events.
- Checkpoint selection by training loss, held-out predictive distortion, and oracle downstream validation.

E6. Statistical protocol

- Use at least three seeds for decisive matched runs and report confidence intervals or paired bootstrap intervals for downstream differences.
- Preregister the primary metric per dataset and the allowed hyperparameter search budget.
- Declare a smallest practically meaningful effect before training; avoid interpreting sub-variance wins as mechanism confirmation.
- Report Pareto fronts over quality, training FLOPs, target-system FLOPs, wall-clock time, memory, and inference cost.

10. Risks, mitigation, and kill tests

Risk	Why it matters	Mitigation or kill test
Predictive semantics are wrong	Spurious context cues may be more predictable than task-relevant attributes.	Domain-held-out targets, nuisance randomization, context interventions, background controls, and channel deletion tests.
Frozen tokenizer is too weak	The context law is measured through an impoverished descriptor.	Reconstruction and attribute audits; tokenizer capacity sweeps; multiple independent tokenizers.
Frozen tokenizer is too semantic	The method becomes teacher distillation rather than a new predictive principle.	Target-only transfer, teacher-size controls, information-preserving targets, and random or low-level tokenizer comparisons.
Conditional mean is not learned	The network memorizes one context per event.	Repeated-event synthetic tests, stronger held-out context splits, target resampling, reduced capacity, and regularization sweeps.
Observation leakage	The event token sees withheld context directly.	Hard masks, causal attention restrictions, target-region interventions, and leakage classifiers.
Conditional target is too noisy	One realized context creates high stochastic gradient variance.	Target averaging, robust loss, larger batches, curriculum, and predictability-based channel weights.
Position shortcuts	Dataset framing and absolute coordinates dominate prediction.	Relative coordinates, coordinate dropout, randomized crops, and within-position cross-image tests.
Kernel mismatch	Embeddings become nearly constant or nearly orthogonal.	Multi-bandwidth blocks, spectrum inspection, median-distance initialization, and held-out neighborhood diagnostics.
Token capacity bottleneck	Direct RFF regression consumes dimensions needed for transfer.	Dimension allocation sweeps, fixed compression, intermediate-layer targets, and comparison with a learned projector.
Transport conflict	Semantic tokens legitimately change when evidence or context changes.	Restrict to compatible overlap-valid maps; recompute event channels; anneal λ_T using residual diagnostics.
Compute overhead	Frozen context evaluation erases objective gains.	Caching, small target encoders, precomputed crop templates, and full-system FLOP/wall-clock reporting.
Metric does not predict transfer	The proposed theory is descriptive but not useful for model selection.	Frozen-checkpoint zoo and preregistered checkpoint-ranking tests before large-scale training.

10.1. Single weakest assumption

Weakest assumption

For mainstream visual tasks, the distinctions that matter are sufficiently reflected in the conditional distributions of practical frozen context descriptors under a graded foveal event channel, and those predictive states admit a useful finite-dimensional realization that can be learned without context leakage or memorization.

10.2. Progression gate for scale-up

Scale-up gate

Proceed from ViT-S to ViT-B only if PIVOT matches global transfer within ordinary run variance, produces a clear dense, retrieval, robustness, or calibration advantage, and its held-out predictive metrics rank checkpoints more reliably than baseline SSL losses. Proceed toward DINOv2/v3-scale claims only after the mechanism survives full-system compute accounting and mechanism-separating ablations.

11. Novelty position

Every component has clear ancestry: predictive coding and JEPA use context as a self-supervised signal; kernel mean embeddings represent probability laws in Hilbert space; random Fourier features make shift-invariant kernels finite; DINO provides strong view-based relational learning; dense correspondence methods enforce local alignment; and redundancy-reduction methods protect noncollapse. The contribution is the division of labor and the direct deployed-token property.

Most defensible novelty statement

PIVOT trains a retained dense visual token to approximate the conditional mean embedding of independently frozen, multichannel context features under a declared foveal event channel. Fixed characteristic random features make a finite predictive-kernel sketch explicit and positive semidefinite; fixed pooling induces coarse geometry; and verified known-warp transport supplies a separate spatial coordinate constraint.

The claim is not that probabilistic representations, latent prediction, Gram geometry, transport, random features, or anti-collapse regularization are individually new. The claim is that this combination turns predictive equivalence into a direct, measurable, scalable hypothesis about the deployed token field without declaring a universal marginal distribution for representations.

12. Research success criteria and conclusion

12.1. Primary success criteria

18. A compute-matched win over original DINO on at least one preregistered global metric and one dense, retrieval, robustness, or calibration metric, without a material regression across the remaining core suite.
19. Held-out predictive distortion that ranks checkpoints and ablations more reliably than baseline SSL training losses.
20. Evidence that graded foveal context and conditional ambiguity, rather than target-tokenizer distillation or marginal regularization, explain the gain.
21. Benefits that remain after counting target-tokenizer pretraining and target-forward cost.
22. A clear failure boundary on context-silent labels and a transparent account of when DINO-style invariance is superior.

12.2. Bounded conclusion

PIVOT is a credible research direction because its central population optimum is simple, its deployed-token geometry is explicit, and its failure can be measured before frontier-scale training. It directly addresses two conceptual weaknesses in neighboring work: anti-collapse structure is not mistaken for semantic content, and local-to-global prediction is replaced by a declared conditional target that can express ambiguity. The theory is not sufficient on its

own. Its practical validity depends on observation-channel fidelity, target semantics, finite-sample generalization, sketch dimension, optimization, and honest compute accounting.

The first implementation should therefore be deliberately conservative: an ordinary ViT, hard event-channel masking, an independently audited target tokenizer, token-dimensional random-feature blocks, direct regression, overlap-aware transport, and a temporary variance safeguard. The proposal earns scale only if its own held-out metrics predict transfer and if it produces a reproducible Pareto improvement over matched baselines.

12.3. Final backbone-variable check

The defining quantity is the retained token itself. At the population optimum of the direct regression loss, it equals a finite sketch of the conditional mean embedding. The target tokenizer, random features, masks, and warp maps are frozen. No learned projector or predictor can absorb the defining property. Fixed pooling transfers token geometry to regions and images, but its usefulness remains empirical. The remaining scientific risks are target misspecification, finite-dimensional approximation, observation leakage, finite-sample generalization, and optimization - precisely the risks addressed by the proposed pilots and refutation criteria.

13. References

- [1] Caron, M. et al. Emerging Properties in Self-Supervised Vision Transformers. ICCV, 2021.
- [2] Oquab, M. et al. DINOv2: Learning Robust Visual Features without Supervision. Transactions on Machine Learning Research, 2024.
- [3] Simeoni, O. et al. DINOv3. arXiv:2508.10104, 2025.
- [4] Assran, M. et al. Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. CVPR, 2023.
- [5] Bardes, A., Ponce, J., and LeCun, Y. VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. ICLR, 2022.
- [6] Balestriero, R. and LeCun, Y. LeJEP: Provable and Scalable Self-Supervised Learning Without the Heuristics. arXiv:2511.08544, 2025.
- [7] He, K. et al. Masked Autoencoders Are Scalable Vision Learners. CVPR, 2022.
- [8] Zbontar, J. et al. Barlow Twins: Self-Supervised Learning via Redundancy Reduction. ICML, 2021.
- [9] Smola, A., Gretton, A., Song, L., and Scholkopf, B. A Hilbert Space Embedding for Distributions. ALT, 2007.
- [10] Song, L. et al. Hilbert Space Embeddings of Conditional Distributions with Applications to Dynamical Systems. ICML, 2009.
- [11] Rahimi, A. and Recht, B. Random Features for Large-Scale Kernel Machines. NeurIPS, 2007.
- [12] HaoChen, J. Z. et al. Provable Guarantees for Self-Supervised Deep Learning with Spectral Contrastive Loss. NeurIPS, 2021.
- [13] Shalizi, C. R. and Crutchfield, J. P. Computational Mechanics: Pattern and Prediction, Structure and Simplicity. Journal of Statistical Physics, 2001.